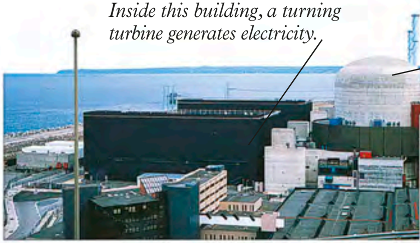
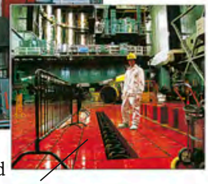


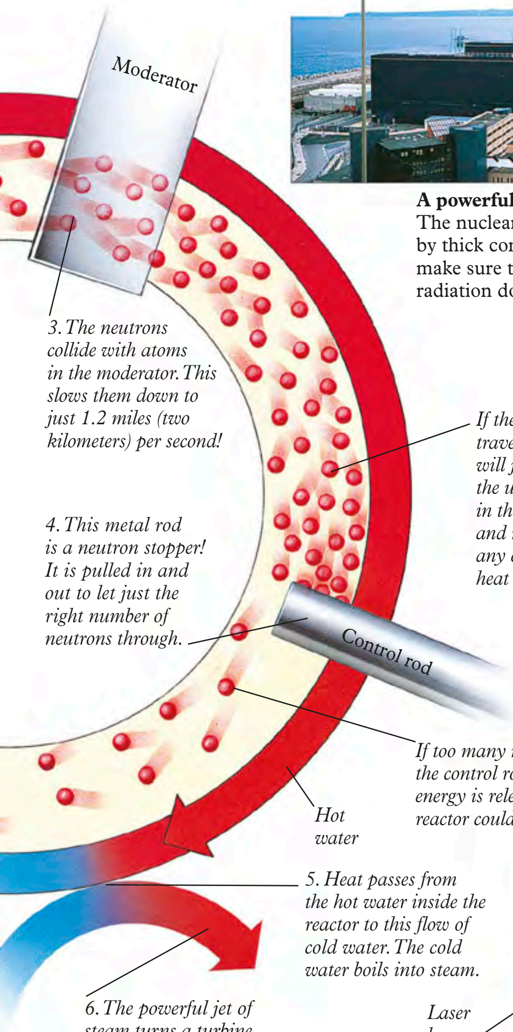
Inside this building, a turning turbine generates electricity.

The nuclear reactor is in here.



The reactor is under this red steel floor.

A powerful building
The nuclear reactor is surrounded by thick concrete walls. These make sure that dangerous radiation does not escape.



3. The neutrons collide with atoms in the moderator. This slows them down to just 1.2 miles (two kilometers) per second!

If the neutrons travel too fast, they will just whizz past the uranium atoms in the fuel rod, and not release any energy to heat up the water.

4. This metal rod is a neutron stopper! It is pulled in and out to let just the right number of neutrons through.

If too many neutrons pass the control rod, too much energy is released, and the reactor could explode.

Hot water

5. Heat passes from the hot water inside the reactor to this flow of cold water. The cold water boils into steam.

6. The powerful jet of steam turns a turbine to generate electricity.

Safe deposit?
A typical nuclear power station produces about 20 bathfuls of very dangerous radioactive waste each year. It is made into a sort of glass and poured into steel tanks, which are coated in concrete and buried.

Less dangerous waste is buried in barrels.

Laser beam

Sunny future
When superhot atoms collide, they fuse together and set energy free. It is this fusion that makes the sun shine. Scientists are trying to build "suns" on Earth by firing lasers at atoms.